

Keywords to consider, research approach, select the best tool, out of the concepts, which is end, a big research question and sub questions:

e.g., Drone control frequently might be influenced by Machine frequency. Refinery in the desert.

Process unitral

Background information: mapping, links to my

1. all list of things,,,

2.

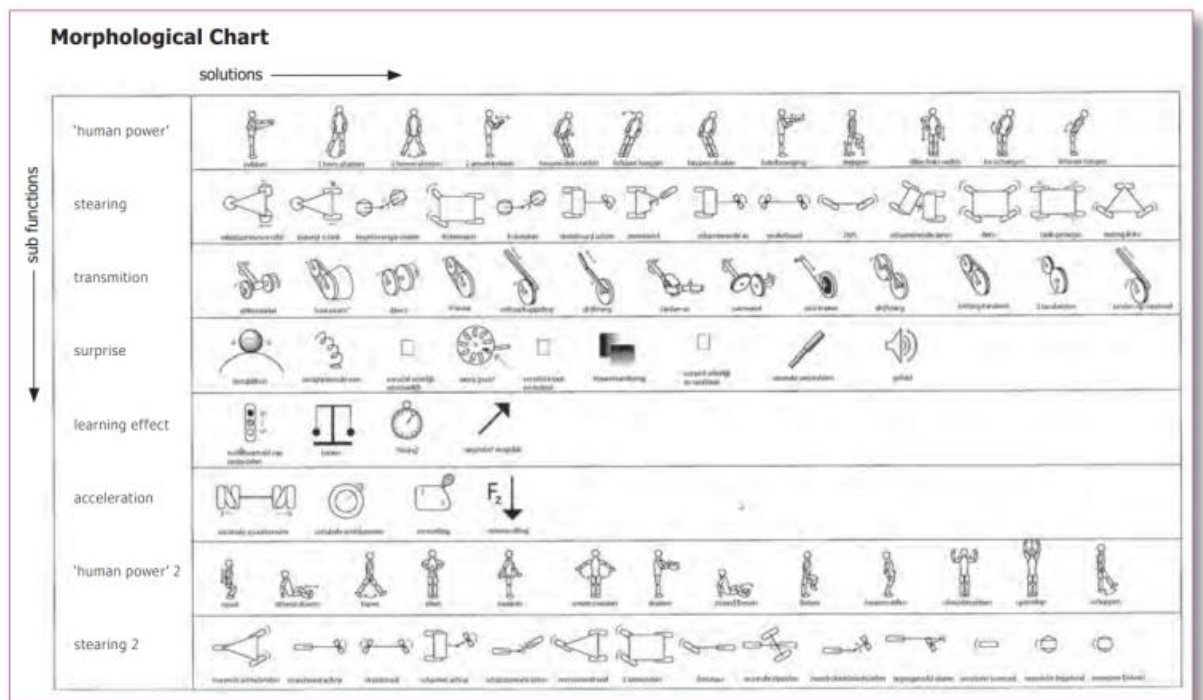
Scope is narrowed.... Shaped on the paper...

Optimal..

morphological diagram

Learning , poorya's paper:

SOTA:



Keywords:

Factory **independency**, (RTLS is out, navigation come)

Fast, reliability, scalability, not only for learning factory, flexibility

Actually problem:

Currently factory get insight I, know what 's happening, those KPI in factory, regular inspection is necessary, common solutions,

Costly, window lock, security issue; like those manual, disturb the production, security is the issue.

How to start a literature review?

Why you think this paper is relevant?

Search platform, **google scholar**, scopus? 99% is not relevant.

Read last paragraph of introduction:

- Conclusion,
- Methods,

Come with **5-10** papers, :

How to select the?

Not limit your, not direct your mind,,, academic skill,

Question:

Why dog?

Why SLAM?

Why dog with SLAM?

Why not NAVIS? Advantage, disadvantage.

They want inspection, but not, develop a autonomous inspection/navigation within the factory, for flexibility, independent of the , need nothing from the factory; fast, reliable, and scalable (to really complex factory)

Many techniques, which is the best for us? Which SLAM packages is better? Key points, SLAM is for me, not to say that.... Comprehensive literature review to say that .

Human, AGV, drone ---- why use the dog, also explain why we use **dog** here?
Interdependency, which combination of technology.

If control in narrow two machine, obstacle, safely navigation,

Metric evaluation, to reach to that navigation, care about safety...

For Fahard: Filter the analysis, dog to select the pathway..... less 人工介入 不用 Wi-Fi, no asking anything,

Solution:

Storyline.

Research plan:

Research on LiDAR,

Leica's technique, inspection in my factory with?

Solutions, plan to

1/06/ Meeting notes from Leicai:

1. why we do this research?

(setup of this devices limitation ...) navigation

Localization (no need for environment) + -> navigation (SLAM map the known environment), where is A and B.

2. navigation strategies

different localization strategies, e.g., tag-based UWB not just localization, also use for navigation,

What would be the best navigation strategies?

Namely why you choose SLAM?

GPS, AI-vision based localization...

But here: **industrial indoor complex environment**, although different tools there,

Visual SLAM, LiDAR SLAM, thermal SLAM?

First, what are the best several SLAM algorithm for ?

Dog, AGV, drone, carrier

3. User interface need to be narrowed down,
4. Validated in Learning factory, Cube basement, and AMC Fraunhofer